King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 208 Major Exam II Term 253

Original exam date: 19 July 2026. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. The general solution of the differential equation $y^{(4)} - y^{(3)} - 3y'' + 5y' - 2y = 0$ is

(a) $y(x) = \left(C_1 + C_2x + C_3x^2\right)e^x + C_4e^{-2x}$

(b) $y(x) = C_1e^x + \left(C_2 + C_3x + C_4x^2\right)e^{-2x}$

(c) $y(x) = (C_1 + C_2x)\,e^x + C_3e^{-x} + C_4e^{-2x}$

(d) $y(x) = \left(C_1 + C_2x + C_3x^2\right)e^x + C_4xe^{-2x}$

(e) $y(x) = (C_1 + C_2x)\,e^x + (C_3 + C_4x)\,e^{-2x}$

2. Given that $y = e^{4x}\cosh x$ is a solution of the differential equation

$$y^{(4)} - 12y^{(3)} + 60y'' - 164y' + 195y = 0,$$

the general solution of the differential equation is

(a) $y = C_1e^{-3x} + C_2e^{-5x} + e^{2x}\,(C_3\cos 4x + C_4\sin 4x)$

(b) $y = C_1e^{5x} + C_2e^{3x} + e^{-2x}\,(C_3\cos 3x + C_4\sin 3x)$

(c) $y = C_1e^{5x} + C_2e^{3x} + e^{2x}\,(C_3\cos 3x + C_4\sin 3x)$

(d) $y = C_1e^{5x} + C_2e^{3x} + e^{x}\,(C_3\cos 5x + C_4\sin 5x)$

(e) $y = C_1e^{-3x} + C_2e^{-5x} + e^{3x}\,(C_3\cos 2x + C_4\sin 2x)$

3. A particular solution of the differential equation $y'' + y' = -e^{-2x}\cos\left(e^{-x}\right)$ is given by $y_p(x) =$

(a) $\cos\left(e^{-x}\right)$

(b) $e^{-x}\cos\left(e^{-x}\right) + \sin\left(e^{-x}\right)$

(c) $\cos\left(e^{-x}\right) - e^{-x}\sin\left(e^{-x}\right)$

(d) $\sin\left(e^{-x}\right)$

(e) $\cos\left(e^{-x}\right) + e^{-x}\sin\left(e^{-x}\right)$

4. If $y(x)$ is the solution of the initial-value problem

$$35y'' - y' - 12y = 0, \qquad y(0) = 5, \quad y'(0) = 3,$$

then $y(1) =$

(a) $3e^{3/5} + 2e^{-4/7}$

(b) 0

(c) $5e^{-4/7}$

(d) $2e^{3/5} + 3e^{-4/7}$

(e) $5e^{3/5}$

5. The rank of the matrix $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 1 & 3 & 4 \end{bmatrix}$ is

(a) 4

(b) 1

(c) 3

(d) 0

(e) 2

6. If $y_p = Axe^{3x} + Be^{3x} + C$ is a particular solution of the differential equation $y'' + 9y = 2xe^{3x} + 5$, then $9A + 27B + 9C =$

(a) 9

(b) 5

(c) -5

(d) 0

(e) 1

7. If the solution space of the system

$$\begin{aligned} x_1 - 3x_2 - 9x_3 - 5x_4 &= 0, \\ 2x_1 + x_2 - 4x_3 + 11x_4 &= 0, \\ x_1 + 3x_2 + 3x_3 + 13x_4 &= 0 \end{aligned}$$

consists of all linear combinations of the vectors $v_1 = (a, b, 1, 0)$ and $v_2 = (c, d, 0, 1)$, then $a+b+c+d =$

(a) -4

(b) 4

(c) -6

(d) 6

(e) 0

8. If $W(x)$ is the Wronskian of the functions

$$f_1(x) = \sin(5x) + 2\cos(5x), \quad \text{and} \quad f_2(x) = 3\sin(5x) + \cos(5x),$$

then $W(x) =$

(a) 5

(b) -25

(c) 25

(d) 0

(e) $25\cos(5x)$

9. A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = (A + Bx)\cos 2x + (C + Dx)\sin 2x + Ee^{2x} + Fe^{-2x}$$

is

(a) $y^{(6)} + 4y^{(4)} - 16y'' - 64y = 0$

(b) $y^{(6)} - 8y^{(4)} + 16y'' = 0$

(c) $y^{(6)} + 8y^{(4)} + 16y'' = 0$

(d) $y^{(4)} - 16y = 0$

(e) $y^{(6)} - 4y^{(4)} - 16y'' + 64y = 0$

10. If the matrix $A = \begin{bmatrix} 6 & -2 \\ 3 & 1 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

(c) $P = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}$

(d) $P = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$

11. Consider the subspace S of $\mathbb{R}^4$ defined by $S = \{(a, b, c, d) \in \mathbb{R}^4 \mid a+2b+3c+4d = 0,\ b+2c+3d = 0\}$. A basis of S consists of the vectors

(a) $v_1 = (1, 2, 1, 0),\ v_2 = (2, -3, 0, 1)$

(b) $v_1 = (-1, -2, 1, 0),\ v_2 = (2, -3, 0, 1)$

(c) $v_1 = (2, -3, 0, 1),\ v_2 = (4, -6, 0, 2)$

(d) $v_1 = (1, -2, 1, 0),\ v_2 = (2, -3, 0, 1)$

(e) $v_1 = (1, -2, 1, 0),\ v_2 = (2, -4, 2, 0)$

12. An eigenvector associated with the eigenvalue $\lambda = 5$ of the matrix $A = \begin{bmatrix} 3 & 1 \\ 4 & 3 \end{bmatrix}$ is $\begin{bmatrix} a \\ 2 \end{bmatrix}$, where $a =$

(a) 2

(b) -1

(c) -2

(d) 1

(e) 0

13. The characteristic equation of the matrix $A = \begin{bmatrix} 0 & 1 & -1 \\ -3 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ is

(a) $\lambda^3 + 4\lambda^2 + 8\lambda - 8 = 0$

(b) $-\lambda^3 + 4\lambda^2 - 8\lambda + 8 = 0$

(c) $-\lambda^3 + 4\lambda^2 + 8\lambda + 8 = 0$

(d) $-\lambda^3 - 4\lambda^2 - 8\lambda + 8 = 0$

(e) $-\lambda^3 + 4\lambda^2 - 6\lambda + 8 = 0$

14. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$y^{(3)} - y'' - 12y' = x - 2xe^{-3x}$$

is

(a) $y_p = x(Ax + B) + xe^{-3x}(Cx + D)$

(b) $y_p = x(Ax + B) + e^{-3x}(Cx + D)$

(c) $y_p = x^2(Ax + B) + xe^{-3x}(Cx + D)$

(d) $y_p = Ax + B + e^{-3x}(Cx + D)$

(e) $y_p = x(Ax + B) + x^2e^{-3x}(Cx + D)$

15. If the characteristic polynomial of the matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 1 & -1 & -2 \end{bmatrix}$ is $p(\lambda) = -(\lambda - 3)^2(\lambda + 2)$, then a basis for the eigenspace of $\lambda = 3$ is $v_1 = \begin{bmatrix} 1 \\ \alpha \\ 0 \end{bmatrix}$, and $v_2 = \begin{bmatrix} \beta \\ 0 \\ 1 \end{bmatrix}$, where $\alpha + \beta =$

(a) 0

(b) -5

(c) 5

(d) 6

(e) -6

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(a)**	$r^4 - r^3 - 3r^2 + 5r - 2 = (r-1)^3(r+2)$: $y = (C_1 + C_2x + C_3x^2)e^x + C_4e^{-2x}$.
2	**(c)**	$e^{4x}\cosh x$ gives $r = 5, 3$; quotient $r^2 - 4r + 13$ gives $r = 2 \pm 3i$.
3	**(a)**	VoP, $y_1 = 1$, $y_2 = e^{-x}$, $W = -e^{-x}$: $u_1 = t\sin t + \cos t$, $u_2 = -\sin t$ ($t = e^{-x}$).
4	**(e)**	$(5r-3)(7r+4) = 0$; $c_1 = 5$, $c_2 = 0$: $y = 5e^{3x/5}$, $y(1) = 5e^{3/5}$.
5	**(e)**	$R_3 = R_1 + R_2$, $R_4 = 2R_1 + R_2$, so two pivots: $\operatorname{rank}(A) = 2$.
6	**(b)**	$18A = 2$, $6A + 18B = 0$, $9C = 5$: $A = \frac{1}{9}$, $B = -\frac{1}{27}$, $C = \frac{5}{9}$, sum $= 5$.
7	**(c)**	RREF: $x_1 = 3x_3 - 4x_4$, $x_2 = -2x_3 - 3x_4$: $v_1 = (3, -2, 1, 0)$, $v_2 = (-4, -3, 0, 1)$, sum -6.
8	**(c)**	$W = f_1f_2' - f_1'f_2 = 25\sin^2 5x + 25\cos^2 5x = 25$.
9	**(a)**	Roots $\pm 2i$ (double), ± 2: $(r^2+4)^2(r^2-4) = r^6 + 4r^4 - 16r^2 - 64$.
10	**(e)**	$\lambda = 4$: $(1, 1)$; $\lambda = 3$: $(2, 3)$; so $P = [\,(1,1)\ (2,3)\,]$, $D = \operatorname{diag}(4, 3)$.
11	**(d)**	$b = -2c - 3d$, $a = c + 2d$: basis $(1, -2, 1, 0)$, $(2, -3, 0, 1)$.
12	**(d)**	$(A - 5I)\mathbf{v} = \mathbf{0}$: $-2a + 2 = 0$, so $a = 1$.
13	**(b)**	$\det(A - \lambda I) = -\lambda(2-\lambda)^2 + 4(2-\lambda) = -\lambda^3 + 4\lambda^2 - 8\lambda + 8$.
14	**(a)**	Roots $0, 4, -3$: both parts overlap once, so $y_p = x(Ax + B) + xe^{-3x}(Cx + D)$.
15	**(d)**	$A - 3I$: $x_1 = x_2 + 5x_3$, basis $(1, 1, 0)$, $(5, 0, 1)$: $\alpha + \beta = 1 + 5 = 6$.